

FEB12001X – Exam 24 October 2025

FEB12001X Applied Microeconomics 25/26 (ESE) · 11 exercises · 60.0 points

Question 1-Personnel #51846738

6 pts · Last updated 9 Oct, 15:45

The profits of plumbing company Bata are described by:

$$[Q - W] \cdot N(W)$$

where Q is productivity per worker (measured in euros), W the salary that Bata pays each employee, and the function $N(W)$ describes the number of people that work at Bata as a function of the salary paid by Bata. The first derivative of the function $N(W)$ is positive.

Text

What is the value of the first derivative of the function $N(W)$ if Bata hires workers in a perfectly competitive labor market? Give a brief explanation for your answer (maximum two sentences).

2 pts · Open · 1/2 Page

Model answer

In a perfectly competitive labor market, the first derivative of the function $N(W)$ is plus infinity. Starting from the market-clearing wage, a small decrease in the wage paid by Bata leads to a complete drop in labor supply. A small increase in the wage paid by Bata leads to a huge increase in the number of workers willing to work at Bata (in fact all of the workers in this labor market).

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The first-order condition for maximum profits is described by:

$$-N(W) + [Q - W] \cdot N'(W) = 0$$

where $N'(W)$ is the first derivative of the function $N(W)$.

Text

Explain using this first-order condition what the marginal benefits and marginal costs of a salary increase are for Bata.

2 pts · Open · 1/2 Page

Model answer

The first term reflects marginal costs. It is equal to N . If the salary increases by 1, salary costs increase by the number of employees N times 1. The second term reflects marginal benefits. If the salary increases by 1, labor supply increases by $N'(W)$. Bata earns an amount on each worker of $(Q-W)$. And so profits increase by $N'(W)$ times $(Q-W)$.

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During the lecture we have rewritten the first-order condition resulting in the following expression:

$$W = \frac{\eta}{1 + \eta} Q$$

where η is the wage-elasticity of labor supply for Bata.

Text

Suppose $Q = 6.000$ euro per month. What level of the monthly salary will Bata set if the wage-elasticity of labor supply for Bata is equal to 2? How much profit does Bata make per worker per month?

2 pts · Open · 1/2 Page

Model answer

Using the formula, it follows that $W = [2 / (1 + 2)] 6.000 = 4.000$ euro per month. Bata makes $6.000 - 4.000 = 2.000$ profit per worker per month.

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Bata makes $6.000 - 4.000 = 2.000$ profit per worker per month. 1 pt

Question 2-Personnel #51850003

7 pts · Last updated 9 Oct, 16:57

Consider an economy where firms compete for workers and there is free entry and exit of firms, so that expected profit of each firm is zero. Workers in this economy differ in productivity. The least productive workers have a productivity of 20 and the most productive workers have a productivity of 80. Workers' productivity is uniformly distributed on the interval 20 to 80. Piece-rate pay is not possible; workers are paid a salary. At the moment salaries are set, the productivity of an individual worker is not known, neither to the individual worker, nor to firms. Workers only care about salary, not about any other job aspects.

Text

How does the distribution of salaries look like in this economy in equilibrium?

2 pts · Open · 3/10 Page

Model answer

Competition between firms makes sure that expected profits are zero and hence salary is equal to expected productivity. The expected value of productivity of a worker is $(20+80)/2 = 50$. All workers earn a salary of 50.

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The expected value of productivity of a worker is $(20+80)/2 = 50$. All workers earn a salary of 50. 1 pt

Suppose now that, before salaries are set, each worker can have his or her productivity measured and put on a certificate, all at no cost. The certificate can be used as a proof of one's productivity when applying at a firm.

Text

How does the distribution of salaries look like in this economy in equilibrium?

2 pts · Open · 9/20 Page

Model answer

Each worker will have his or her productivity measured, because it is costless and possible useful. Each worker who learns that his productivity is above the market salary for workers without a certificate will use the certificate to get a higher wage. Starting at a market salary of 50, this means that all workers with a productivity between 50 and 80 will use the certificate. This in turn means that the market salary for people without a certificate goes down to $(20+50)/2 = 35$, because otherwise firms make a loss. Because of this salary decline, also the workers with a productivity between 35 and 50 will use the certificate, resulting in another decline in the market salary, this time to $(20+35)/2 = 27,5$. And so on and so forth. Until all workers use the certificate (except perhaps the workers at the bottom). All workers will earn their productivity.

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pt

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1
pt

Assume now that each worker already knows his or her productivity before a possible acquisition of a certificate, but that firms do not know a worker's productivity. Workers can buy a certificate that enables them to prove what their productivity is to firms. The cost of the certificate are equal to 20. Workers aim to maximize disposable salary, that is salary minus possible cost of the certificate.

Text

Which workers will buy a certificate in equilibrium?

3 pts · Open · 1/2 Page

Model answer

A worker who buys a certificate earns his or her productivity q . His or her disposable income is equal to $q - 20$, because the certificate costs 20. Workers without a certificate earn the average productivity of all workers who do not buy a certificate. Denoting the productivity of the indifferent worker by q^* , then this is equal to $(q^* + 20)/2$, because a certificate is least attractive for less productive workers. In equilibrium it holds that $q^* - 20 = (q^* + 20)/2$. Solving gives: $q^* = 60$. All workers with productivity 60 and higher will buy a certificate. The salary for people without a certificate is $(20 + 60)/2 = 40$.

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In equilibrium it holds that $q^* - 20 = (q^* + 20)/2$. Solving gives: $q^* = 60$. All workers with productivity 60 and higher will buy a certificate. The salary for people without a certificate is $(20 + 60)/2 = 40$. 1 pt

Question 3-Personnel #50629711

10 pts · Last updated 31 Oct, 13:49

Consider a profit-maximizing firm employing two workers and operating for two periods. In the first period, both workers earn salary W . In the second period, one of them is promoted to a management job and earns salary $B = W + Z$, whereas the other is retained and again earns salary W . Which worker is promoted depends partly on workers' effort and partly on luck. This is represented by the following function:

$$p_i = \frac{1}{2} + \pi(e_i - e_j),$$

where p_i is the probability that worker i is promoted, e_i is worker i 's effort in period 1, and e_j is worker j 's effort in period 1. Each worker's costs of effort is given by $\frac{1}{2} \theta e^2$.

Each worker aims to maximize the sum of his utility in period 1 and period 2. A worker's utility in a period is equal to his income minus his costs of effort and, in addition to this, the worker who has been promoted enjoys an extra utility of amount P , which reflects the utility from feeling proud. There are no costs of effort in period 2. There is also no production in period 2. Each worker chooses his effort in period 1 independently of the other worker. Assume that parameters are such that the firm wishes to employ the employees and that each employee wishes to stay at the firm in period 2 even if he did not get the promotion.

Text

Derive the first-order condition describing the optimal level of effort of worker i in period 1.

2 pts · Open · 3/10 Page

Model answer

The worker's expected utility at the start of period 1 is

$$\left[\frac{1}{2} + \pi(e_i - e_j) \right] (Z + P) + 2W - \frac{1}{2} \theta e_i^2. \text{ Differentiating and setting equal to zero gives:}$$

$$\pi(Z + P) - \theta e = 0$$

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$$\left[\frac{1}{2} + \pi(e_i - e_j) \right] (Z + P) + 2W - \frac{1}{2} \theta e_i^2. \quad 1 \text{ pt}$$

Differentiating and setting equal to zero gives: $\pi(Z + P) - \theta e = 0$

1 pt

Rather than working at this firm, a worker can decide to work somewhere else, and enjoy utility V per period.

Text

Show that the minimum level of the salary W at which each worker is just willing to take the job at this firm is equal to: $V - \frac{1}{4}(Z + P) + \frac{1}{4}\theta \left(\frac{\pi(Z + P)}{\theta} \right)^2$

3 pts · Open · 1/2 Page

Since workers choose the same effort, it holds that $p_i = p_j = \frac{1}{2}$ 1 pt

Expected utility when taking the job thus is

$2W + \frac{1}{2}(Z + P) - \frac{1}{2}\theta e^2$ which must be equal to expected utility when not taking the job, which is equal to $2V$ 1 pt

Substituting the level of effort derived under part a and rewriting gives the expression in the question. 1 pt

It is clear from the expression in question 3b that when the utility from being proud P goes up, the minimum level of the salary W may increase or decrease, see the second and third term of the expression. This reflects two opposing effects of P on the attractiveness of the job. Give an economic interpretation of these two opposing effects. Do not write "math in words".

2 pts · Open · 3/10 Page

When the utility from being proud is higher, the job becomes more attractive because promotion entails a higher utility gain. 1 pt

However, when the utility from being proud is higher, the job becomes less attractive because it makes the worker work harder in period 1, which increases costs of effort. 1 pt

Assume each unit of effort results in one unit of output that is sold at price R . Derive the profit-maximizing level of Z .

3 pts · Open · 9/10 Page

Profits equal $2Re - Z - 4W$ 1 pt

Substituting the effort from 3a and the wage from part 3b gives:

$2R \frac{\pi(Z + P)}{\theta} - Z - 4V + Z + P - \theta \left(\frac{\pi(Z + P)}{\theta} \right)^2$ 1 pt

Take the derivative to Z , set equal to zero and solve gives $Z = \frac{R}{\pi} - P$ 1 pt

Question 4-Personnel #50629712

4 pts · Last updated 9 Oct, 17:08

A worker has the following utility function: $U = V(Y) + X(24 - H)$, where Y is the worker's income per day, H is her number of work hours per day, $V'(Y) > 0$, $V''(Y) < 0$, $X'(24 - H) > 0$, and $X''(24 - H) < 0$. The worker earns an hourly wage equal to W and chooses her work hours so as to maximize her utility. The function $V(Y)$ describes the worker's utility from income. The function $X(24 - H)$ describes the worker's utility from leisure.

Text

The first-order condition for the worker's optimal number of work hours is $WV'(WH) - X'(24 - H) = 0$. Give a brief interpretation for each of the two terms of this first-order condition.

2 pts · Open · 3/10 Page

0 pts

The first term describes the effect of one hour more work on income and hence on the utility from income. (Also correct is: When you work more, you earn a higher income, which makes you more happy)

1
pt

The second term describes the effect of one hour more work on utility from leisure, which decreases. (Also correct is: When you work more, you have less leisure time, which makes you less happy)

1
pt

Use the first-order condition described in question 4a to examine the consequences of an increase in the hourly wage W on the number of work hours chosen by the worker. A discussion in two clear sentences of how an increase in the hourly wage affects the marginal benefits and/or costs of work hours suffices. Do not just mention the name of the effects, but describe them in clear sentences (just name-dropping yields no points).

2 pts · Open · 2/5 Page

An increase in the hourly wage affects the marginal benefits from work hours in two ways: first, an additional work hour now increases income more (or, you can also write: becomes financially more rewarding), which increases optimal work hours.

1
pt

Second, for a given number of hours, the worker's income rises when the hourly wage goes up, which in turn reduces the marginal utility from income, and so reduces optimal work hours.

1
pt

Question 5-Personnel #52037095

3 pts · Last updated 6 Oct, 06:34

The owner of a large retail chain would like to know whether the current remuneration policy is optimal. She asks a consultant to examine which remuneration policies are used by competitors. His analysis shows that large retail chains which use variable pay perform on average less well than large retail chains which do not use variable pay.

Text

Can we conclude from the analysis that the use of variable pay worsens the performance of large retail chains? Explain in one clear sentence why.

1 pt · Open · 1/4 Page

No, because it is just a correlation. Large retail chains with variable pay may differ in other respects 1
from large retail chains without variable pay, which may cause differences in performance. pt

Suppose the owner of the large retail chain decides to run an experiment on the effects of variable pay. Describe two requirements that the experiment needs to satisfy to deliver reliable conclusions on the effects of variable pay in this large retail chain.

2 pts · Open · 1/4 Page

The experiment needs to randomly assign stores to a control group and the group that receives variable pay.

The experiment needs to be sufficiently large scale so as to minimize the role of noise.

Also correct is: the experiment needs to last long enough.

1
pt

Also correct is: there should be no contamination, i.e. the introduction of variable pay should not affect the performance of stores in the control group.

Also correct is: it should be possible to measure performance of stores in a reliable manner.

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Also correct is: the experiment needs to last long enough.

1
pt

Also correct is: there should be no contamination, i.e. the introduction of variable pay should not affect the performance of stores in the control group.

Also correct is: it should be possible to measure performance of stores in a reliable manner.

Question 6-Public #50629714

6 pts · Last updated 9 Oct, 17:20

Consider a society that must divide a total sum of Y euro between two individuals, Agnes and Berta. Utility of Agnes is given by

$$U_A = \sqrt{I_A}$$

and utility of Berta is given by

$$U_B = \beta \sqrt{I_B},$$

where I_A and I_B are the amounts of money that Agnes and Berta receive, respectively, and $\beta > 0$ is a parameter. You can assume that $I_A + I_B = Y$. The social welfare function of this society equals

$$SW = wU_A + U_B,$$

where $w > 0$ is a parameter.

Text

Calculate the amounts of I_A and I_B that maximise social welfare.

2 pts · Open · 1/2 Page

Using Lagrange with Lagrange equation

$$w \sqrt{I_A} + \beta \sqrt{I_B} + \lambda(Y - I_A - I_B)$$

or use that it must hold that $wMU_A = MU_B$ or maximise SW after substituting for $I_A = Y - I_B$.

1
pt

Using Lagrange method yields first-order conditions

$$\frac{w}{2\sqrt{I_A}} - \lambda = 0$$

$$\frac{\beta}{2\sqrt{I_B}} - \lambda = 0$$

$$Y - I_A - I_B = 0$$

1
pt

This yields $\frac{w}{2\sqrt{I_A}} = \frac{\beta}{2\sqrt{I_B}}$ such that $I_B = \frac{\beta^2}{w^2} I_A$. Combining with $I_B = Y - I_A$ yields $I_A = \frac{1}{1 + \frac{\beta^2}{w^2}} Y = \frac{w^2}{w^2 + \beta^2} Y$ and $I_B = \frac{\beta^2}{w^2 + \beta^2} Y$.

The amount of I_A that maximises social welfare depends positively on w . Explain why (so: why is I_A optimally higher if w is higher?). Use maximally two sentences.

2 pts · Open · 3/10 Page

A higher w implies that the weight on the utility of A in the social welfare function is higher (relative to the weight of B)

1
pt

This (higher weight) makes that society values (re)distributing to A more.

1 pt

Now suppose that the government is involved in this (re)distribution. In the absence of redistribution, Agnes would obtain the complete amount Y and Berta would obtain 0. Thus, without redistribution $I_A = Y$ and $I_B = 0$. Redistribution by the government leads to a loss in efficiency. This efficiency loss increases in the amount of money redistributed.

Text

Compared to the amount of I_B as derived under question a), will Berta now receive more, less, or the same amount? Explain your answer in a few sentences.

2 pts · Open · 7/20 Page

Redistribution (to Berta) is now increasingly costly as it leads to a loss in efficiency (trade-off- 1
between efficiency and distribution). pt

As a result, it is now optimal to redistribute less than would be socially optimal in the absence of 1
this efficiency loss / trade-off. pt

Question 7-Public #50629715

8 pts · Last updated 9 Oct, 17:10

Suppose there are two individuals, Christina and Dikra, who both derive utility from the consumption of two goods, good x and good y . Christina's utility function is given by

$$U_C = 2x_C + 32\sqrt{y_C},$$

where x_C is the number of units of good x that Christina consumes and y_C is the number of units of good y that Christina consumes.

Text

Show that Christina's marginal rate of substitution MRS_C can be written as

$$MRS_C = \frac{8}{\sqrt{y_C}}.$$

2 pts · Open · 2/5 Page

$$MRS_C = \frac{MU_y}{MU_x}.$$

1 pt

$$MU_x = 2 \text{ and } MU_y = \frac{16}{\sqrt{y_C}}. \text{ Thus } MRS_C = \frac{\frac{16}{\sqrt{y_C}}}{2} = \frac{8}{\sqrt{y_C}}.$$

1 pt

We can write Dikra's marginal rate of substitution as

$$MRS_D = \frac{4}{\sqrt{y_D}},$$

where y_D is the number of units of good y that Dikra consumes. The price of good x equals $p_x = 1$. The price of good y equals $p_y = 4$. Both prices reflect marginal production cost, such that we can write the marginal rate of transformation MRT as

$$MRT = 4.$$

We assume that Christina and Dikra have sufficiently high incomes, such that they both consume good x and good y . Throughout, assume that good x is a private good.

Text

Suppose good y is also a private good. Derive the socially efficient levels of consumption of good y for both Christina and Dikra.

2 pts · Open · 2/5 Page

$$MRS_C = MRT \text{ and } MRS_D = MRT$$

1 pt

Substituting for MRS and MRT yields $\frac{8}{\sqrt{y_C}} = 4$ such that $y_C = 4$ and $\frac{4}{\sqrt{y_D}} = 4$ such that $y_D = 1$.

1 pt

Suppose good y is a public good. Derive the socially efficient levels of consumption of good y for both Christina and Dikra.

2 pts · Open · 1/2 Page

$$MRS_C + MRS_D = MRT$$

1 pt

Public good implies that they consume the same units: $y_C = y_D = y$. Substituting yields

$$\frac{8}{\sqrt{y}} + \frac{4}{\sqrt{y}} = 4 \text{ such that } y = 9.$$

1
pt

Suppose good y is a public good and that Christina and Dikra decide independently how many units of good y to provide. Determine the Nash equilibrium.

2 pts · Open · 2/5 Page

Dikra provides 0 units (free-rides) in the Nash-equilibrium, (as Christina provides more than 1 unit and $MRS_D < 4$ for $y > 1$). 1 pt

Given that Dikra provides 0 units, Christina provides the number of units that maximise her own utility: $\frac{8}{\sqrt{y_C}} = 4$ such that $y_C = 4$ 1 pt

Question 8-Public #52159474

4 pts · Last updated 9 Oct, 17:14

Suppose that the northern part of a country contains natural resources. The government decides to extract these resources and sell them on the world market. The revenues benefit all citizens of the country equally. The extraction of the natural resources hurts the citizens that live in the northern part of the country. The marginal cost of the extraction to the citizens in the north increase in the level of resource extraction. The government is fully aware of all benefits and cost of the resource extraction. It decides to extract more than the socially efficient level of resources.

Text

Give a normative explanation for why the government decides to extract more than the socially efficient level of resources. Use maximally two sentences.

2 pts · Open · 7/20 Page

Distributional arguments, such as:

The people in the northern part of the country are relatively rich. The government maximises a social welfare function and therefore cares about redistribution from the north to the rest.

2

or

pts

The government maximises a social welfare function which gives relatively low weights to people in this part of the country.

Efficiency arguments (including any benefits of additional government expenditures as well as a possible reduction of the use of distortionary taxation) are incorrect here, as the government takes the benefits of the revenues into account.

0

pts

Give a positive explanation for why the government decides to extract more than the socially efficient level of resources. Use maximally two sentences.

2 pts · Open · 7/25 Page

Anything related to (the process of) collective decision-making, such as

The citizens in the northern part of the country are a minority. The majority votes for politicians that promise (inefficiently) high levels of extraction.

2

or

pts

Politicians are corrupt, and receive part of the revenues (for instance as bribes).

Question 9 - Public #52160450

9 pts · Last updated 9 Oct, 17:14

Consider the perfectly competitive market for good G . Demand can be given by

$$P = 200 - 2Q$$

and supply can be given by

$$P = 20 + Q,$$

where P is the price per unit of good G in euro and Q is the quantity of good G .

Text

Determine the Excess Burden of a unit tax of 30 euro per unit of good G , levied on the producers of good G .

2 pts · Open · 12/25 Page

Pre-tax equilibrium: $200 - 2Q = 20 + Q$ gives $Q = 60$. Post-tax equilibrium:

1

$200 - 2Q = 20 + Q + 30$ gives $Q = 50$.

pt

Excess Burden equals $(1/2) * 30 * (60-50) = 150$ euro.

1 pt

In this exercise, consumers bear share $\frac{2}{3}$ of the tax burden. Would consumers bear a larger or smaller share of the tax burden if demand was less elastic? Explain your answer in a few sentences.

2 pts · Open · 3/10 Page

A less elastic demand implies that alternatives for the consumption of good G are less valuable. 1 pt

This implies that consumers will substitute away less after the introduction of a tax, such that (the 1 consumer price will increase more and) they bear a larger share of the tax burden. pt

Suppose the consumption of good G leads to external cost. The marginal external cost can be represented by $MEC = \alpha Q$, where $\alpha > 0$ is a parameter.

Text

Show that the Pigouvian tax on good G is equal to $\frac{180\alpha}{3 + \alpha}$.

2 pts · Open · 1/2 Page

Pigouvian tax is equal to the MEC at the socially efficient level of consumption. 1 pt

Socially efficient level of consumption is given by $200 - 2Q = 20 + Q + \alpha Q$, which can be solved for $Q = \frac{180}{3 + \alpha}$. Substituting in MEC yields that the Pigouvian tax equals $\frac{180\alpha}{3 + \alpha}$. 1 pt

Suppose that it is not possible for the government to impose a tax on good G . Instead, the government considers a ban on good G , which would result in $Q = 0$. For which values of α does a ban on good G lead to higher social efficiency compared to the situation without government intervention in the market for good G ?

3 pts · Open · 3/5 Page

A ban implies no consumer surplus, no producer surplus, and no external cost. 1 pt

Market outcome $Q = 60$ yields consumer surplus $CS = (1/2) * 60 * (200-80) = 3600$, producer surplus $PS = (1/2) * 60 * (80-20) = 1800$, and external cost $(1/2) * 60 * 60\alpha = 1800\alpha$. 1

Or: integrate $[200 - 2Q - (20 + Q + \alpha Q)]$ with respect to Q and substitute for $Q = 60$ to get $5400 - 1800\alpha$ pt

Solving for $3600 + 1800 - 1800\alpha \leq 0$ yields $\alpha \geq 3$ 1 pt

Question 10-Public #52177621

3 pts · Last updated 9 Oct, 17:27

Consider a country with three groups, Rich, Middle class, and Poor. Within each group, citizens are identical. Each citizen maximises its own utility and consumes all income. All citizens have diminishing marginal utility of leisure and of consumption. All citizens choose how many hours to work. Middle class has a lower income than Rich and a higher income than Poor. The three groups consume different goods: Rich consumes only good R , Middle class only consumes good M , and Poor only consumes good P . The supply of all goods is perfectly elastic. The elasticity of demand for good R is equal to 3, the elasticity of demand for good M is equal to 1, and the elasticity of demand for good P is equal to $\frac{1}{3}$. The government raises revenue using a tax system that imposes ad valorem taxes on consumption. The tax rate on good R is t_R , the tax rate on good M is t_M , and the tax rate on good P is t_P . There are no other taxes possible. Suppose the government wants to minimise the total excess burden of taxation.

Text

Which relations between the different ad valorem tax rates must hold in the optimal tax system for this country?

2 pts · Open · 2/5 Page

To minimize the totale excess burden of taxation, the Ramsey rule / inverse elasticity rule must hold. 1 pt

Thus, it must hold that $\frac{t_R}{t_M} = \frac{1}{3}$, $\frac{t_R}{t_P} = \frac{1}{9}$, and $\frac{t_M}{t_P} = \frac{1}{3}$. Or, equivalently, that

$$t_R = \frac{1}{3} t_M = \frac{1}{9} t_P. \quad 1$$

Someone who did not use the Ramsey rule (and thus did not get the first point) can get this point pt if the answer is given that $t_R < t_M < t_P$ with the explanation that the Excess Burden increases in the elasticity of demand.

Is this optimal tax system progressive, proportional, or regressive? Explain your answer in one sentence.

1 pt · Open · 1/4 Page

This tax system is regressive, as the average tax rate decreases in income: $t_R < t_M < t_P$. 1 pt

Additional answer space #50629716

0 pts · Last updated 8 Oct, 17:26

Only use this extra space if you made a mistake in one of your answers. Please do not forget to **indicate in the answer box of that particular question where the correct answer can be found.**

Otherwise, the answers in the additional space below will not be reviewed.

NB: Do **not** use this additional space for answers that are too long.

Text

Additional answer box 1

Open · 4/5 Page

Additional answer box 2

Open · 1/2 Page

Additional answer box 3

Open · 1/2 Page

Additional answer box 4

Open · 1/2 Page

Additional answer box 5

Open · 1/2 Page